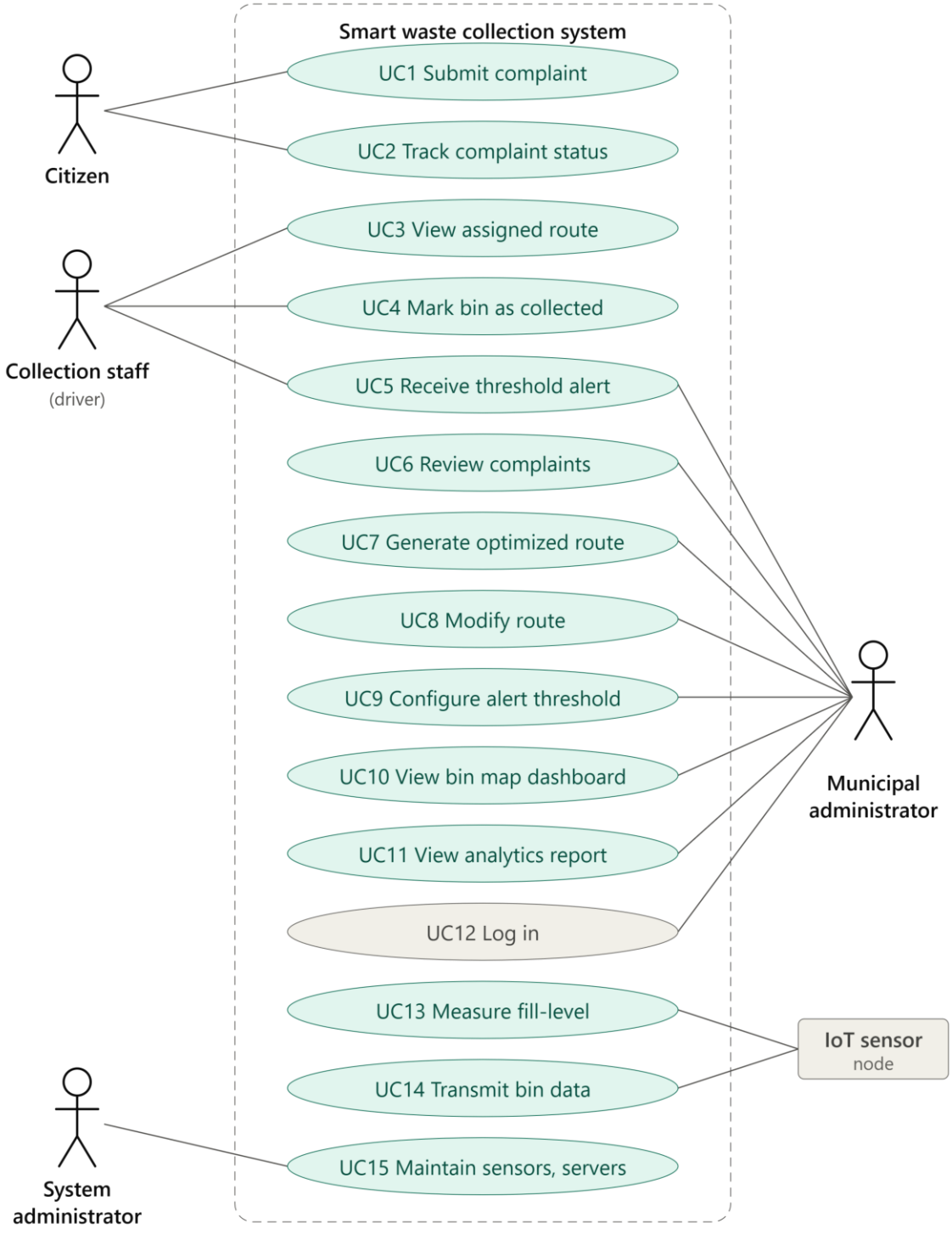
USE CASE DIAGRAM



Actors and their roles

Actor	Role in the system
Citizen	Reports overflowing bins or requests an unscheduled pickup through the app or web form. Tracks the status of their complaint.
Collection staff / driver	Receives the day's optimized route and bin alerts on the mobile app. Marks bins as collected after pickup.
Municipal administrator	Controls the system from the web dashboard. Sets thresholds, generates and adjusts routes, monitors bins, reviews complaints and reads analytics.
System administrator	Technical maintainer. Keeps servers, sensors and connectivity running.
IoT sensor node	Secondary (non-human) actor. The ultrasonic sensor and microcontroller on each bin measure fill-level and send it to the backend.

Use case specifications

ID / Use case	Actor	Function	Precondition	Postcondition
UC1 Submit complaint	Citizen	Report an overflowing bin or request a pickup, with auto-detected location and an optional photo (REQ-CF-1, CF-2).	App or web form is open; location is available or selectable.	Complaint is stored with a unique tracking ID and appears on the admin dashboard as "Pending".
UC2 Track complaint status	Citizen	View whether a complaint is Pending or	Complaint exists and the citizen	Current status is displayed; no data changes.

ID / Use case	Actor	Function	Precondition	Postcondition
		Resolved (REQ-CF-3).	has its tracking ID.	
UC3 View assigned route	Driver	See the day's optimized route as a list or map.	Driver is logged in and a route has been generated and pushed.	Route is displayed on the driver's device.
UC4 Mark bin as collected	Driver	Confirm that a bin on the route has been emptied.	Bin is on the driver's assigned route and has been physically emptied.	Bin is recorded as collected and its fill-level resets. Linked complaints are auto-marked "Resolved".
UC5 Receive threshold alert	Driver, Administrator	Get a push notification when a bin crosses its threshold (REQ-AL-2, AL-3).	Threshold is configured and the sensor has sent a reading above it.	Alert is delivered within 5 minutes and logged with bin ID and timestamp.
UC6 Review complaints	Administrator	Look at citizen complaints forwarded to the dashboard and act on them.	Admin is authenticated and at least one complaint exists.	Complaint is reviewed, and any follow-up pickup is arranged.
UC7 Generate optimized route	Administrator	Compute the shortest route over all bins at or above the collection	Admin is authenticated; bin locations are loaded; at least one bin is above	Route is shown on the dashboard and pushed to the driver app, within

ID / Use case	Actor	Function	Precondition	Postcondition
		threshold (REQ-RO-1, RO-2).	threshold; the mapping API is reachable.	10 seconds for up to 200 bins.
UC8 Modify route	Administrator	Manually add or remove a bin on a generated route (REQ-RO-3).	A route has already been generated (UC7).	Updated route is saved and pushed to the driver.
UC9 Configure alert threshold	Administrator	Set the alert percentage per bin or globally (REQ-AL-1).	Admin is authenticated.	New threshold is saved and applies to all subsequent readings.
UC10 View bin map dashboard	Administrator	See all bins on a map, colour-coded green, yellow or red by fill-level (REQ-DB-1).	Admin is authenticated; sensors have reported data.	Live map is displayed, and the admin can open a bin's history.
UC11 View analytics report	Administrator	See collection counts and average fill-level per bin over a date range (REQ-DB-2).	Admin is authenticated; historical data exists for the chosen range.	Report is displayed; no data changes.
UC12 Log in	Administrator (and other app users)	Authenticate with username and password, with optional	Account exists.	A valid session is created, or access is denied.

ID / Use case	Actor	Function	Precondition	Postcondition
		OTP (REQ-DB-3).		
UC13 Measure fill-level	IoT sensor node	Measure distance to the waste surface and convert it to a fill percentage at a set interval, default 15 minutes (REQ-BM-1, BM-3).	Sensor is powered (battery or solar) and mounted on the bin.	Fill percentage is ready to send. A failed read is retried and logged.
UC14 Transmit bin data	IoT sensor node	Send bin ID, fill-level and timestamp to the backend over Wi-Fi/GPRS using HTTP or MQTT (REQ-BM-2).	UC13 has produced a reading; network connectivity is available.	Backend stores the reading and updates the bin's dashboard status.
UC15 Maintain sensors and servers	System administrator	Keep hardware, servers and connectivity healthy.	Sysadmin has access to servers and devices.	Sensors and backend are operational and any faults are resolved.